Workshop Sheet - Week 4

Rings and Modules. Here is the Lean syntax for introducing a commutative ring A, an ideal I of A, and an A-module M.

`variable [CommRing A]`	Equip A with a commutative ring structure.
`variable (I : Ideal A)`	Introduce an ideal I of A.
`variable (M : Type*)`	Introduce the underlying type M.
`variable [AddCommGroup M]`	Equip M with an additive commutative group structure.
`variable [Module A M]`	Give M compatible scalar multiplication by A.

1. *Rings.*
   (a) Prove x is a unit if and only if there exists $y \in A$ with $xy = 1$.
   (b) If $x^2 = 0$, use (a) to prove that $1 + x$ is a unit, with inverse $1 - x$.
   (c) Prove that inverses are unique: if $xy = 1$ and $xz = 1$, then $y = z$.
   (d) Use (a) to prove that the product of two units is a unit.
   (e) Let $x, y \in A$ and $m, n \in \mathbb{N}$. If $x^m = 0$ and $y^n = 0$, prove $(x + y)^{m+n} = 0$.
   (f) Deduce that the sum of two nilpotent elements is nilpotent.

2. *Prime ideals.* An ideal $\mathfrak{p}$ is *prime* if $\mathfrak{p} \neq A$ and $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$.
   (a) Consider a map $f : A \to B$, let J be an ideal of B. Show $f^{-1}(J) = \{a : f(a) \in J\}$ is an ideal.
   (b) If $\mathfrak{p}$ is a prime ideal of B, prove that $f^{-1}(\mathfrak{p})$ is prime.
   (c) Prove the *nilradical* is $\mathfrak{N} = \{a \in A : a^n = 0 \text{ for some } n \in \mathbb{N}\}$ an ideal. Hint: Question 1(f).
   (d) Challenge: Prove $\mathfrak{N}$ is the intersection of all prime ideals of A. (Hint: Use *Zorn's lemma*)

3. *Modules.* Let M be an A-module and fix $m \in M$.
   (a) Show that $Am = \{rm : r \in A\}$ is a submodule of M.
   (b) Show that $\mathrm{Ann}(m) = \{r \in A : rm = 0\}$ is an A-submodule of A.
   (c) If $m \neq 0$, prove that $\mathrm{Ann}(m) \neq A$.
   (d) Suppose A is a field and $m \neq 0$. Prove that $\mathrm{Ann}(m) = (0)$.
   (e) For an A-module M, let $\mathrm{Ann}(M) = \{a \in A | am = 0 \text{ for every } m \in M\}$. Prove that

$$\mathrm{Ann}(M) = \bigcap_{m \in M} \mathrm{Ann}(m).$$

4. *Vectors and matrices.* Work in $\mathbb{Q}^2$, represented by `Fin 2 → ℚ` in Lean[1]. Write vectors as `![x, y]` and matrices as `!![1, 2; 0, 1]`. Matrix-vector multiplication is `D *ᵥ v`. Let

$$D = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad v = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

   (a) Compute Dv in Lean using `#eval D *ᵥ v`.
   (b) Evaluate the determinant $\det(D)$ using `#eval D.det`.
   (c) Compute the inverse matrix D^{-1} using `#eval D⁻¹`.

[1] See Mathlib's documentation on determinants and adjugates.